REAL ANALYSIS – EXAMINATION QUESTIONS

Instruction: Answer all questions. Each question carries 15 marks.

Question 1 (15 marks)

(a) Define a bounded set in $\mathbb{R}$. (5 marks)

(b) Define the infimum of a subset of $\mathbb{R}$. (5 marks)

(c) Give one example of a bounded set and determine its infimum. (5 marks)

Question 2 (15 marks)

(a) State the definition of the infimum of a set. (5 marks)

(b) Consider the sequence $G_n = \frac{n}{2n+1}$, $n = 1, 2, 3, \ldots$. Show that it is bounded. (5 marks)

(c) Prove that $\inf G_n = \frac{1}{3}$. (5 marks)

Question 3 (15 marks)

(a) Define supremum and infimum of a set. (5 marks)

(b) Let $G \subset \mathbb{R}$ be bounded. State the relationship between $\sup(-G)$ and $\inf G$. (5 marks)

(c) Prove that $\sup(-G) = -\inf G$. (5 marks)

Question 4 (15 marks)

(a) Explain the concept of union of sets. (5 marks)

(b) Consider the intervals $\left(0, \frac{n}{2n+1}\right)$. Describe their behavior as $n \to \infty$. (5 marks)

(c) Prove that

$$\bigcup_{n=1}^{\infty} \left(0, \frac{n}{2n+1}\right) = \left(0, \frac{1}{2}\right).$$

(5 marks)

Question 5 (15 marks)

(a) Define monotone non-decreasing and monotone non-increasing sequences. (5 marks)

(b) Determine whether the sequence $a_n = \frac{1}{n}$ is monotone and justify your answer. (5 marks)

(c) Classify the following sequences as monotone non-decreasing, monotone non-increasing, or non-monotone:

$$b_n = n^3, \quad c_n = (-1)^n, \quad d_n = \frac{1}{n^2}.$$

(5 marks)

Question 6 (15 marks)

(a) State Bernoulli's Inequality. (5 marks)

(b) Prove Bernoulli's Inequality using mathematical induction. (5 marks)

(c) Use the inequality to estimate a lower bound for $(1+x)^n$ where $x > -1$ and $n \in \mathbb{N}$. (5 marks)